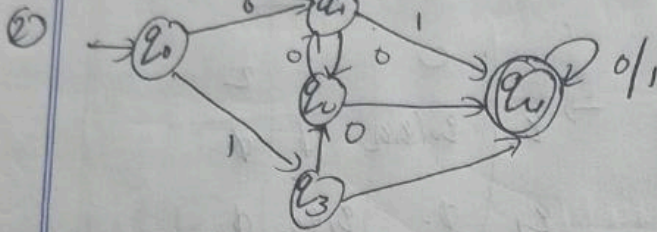
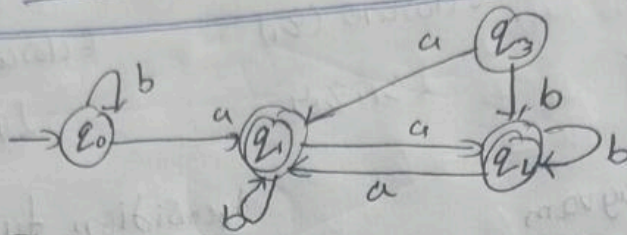


6/07/24

Assignment - 3

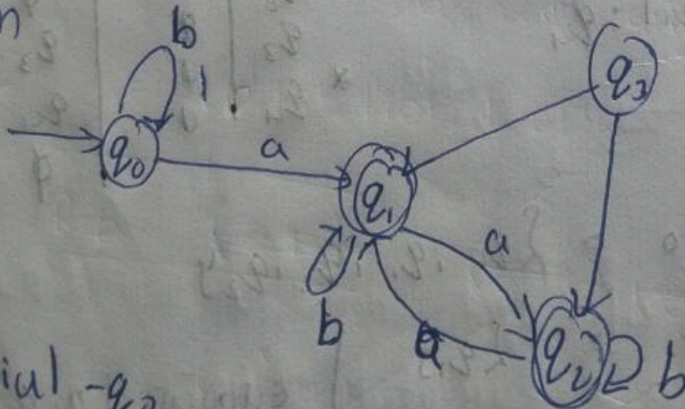
① find min DFA



③ write regular Expression following string the set of all strings length 2 over the A's and B's.

④ write regular Expression string start with 'ab' and with '01'.

① given



initial - q0

final - (q1, q2)

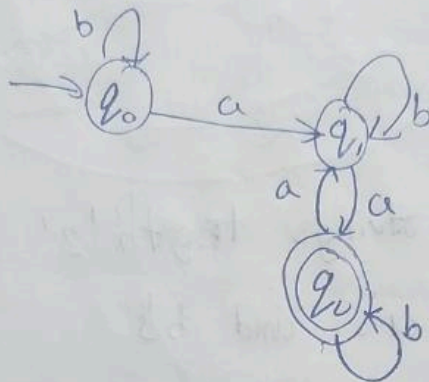
q3 is not reachable state

	a	b
→ q ₀	q ₁	q ₀
q ₁	q ₁	q ₁
*q ₂	q ₁	q ₂
q ₃	q ₁	q ₂

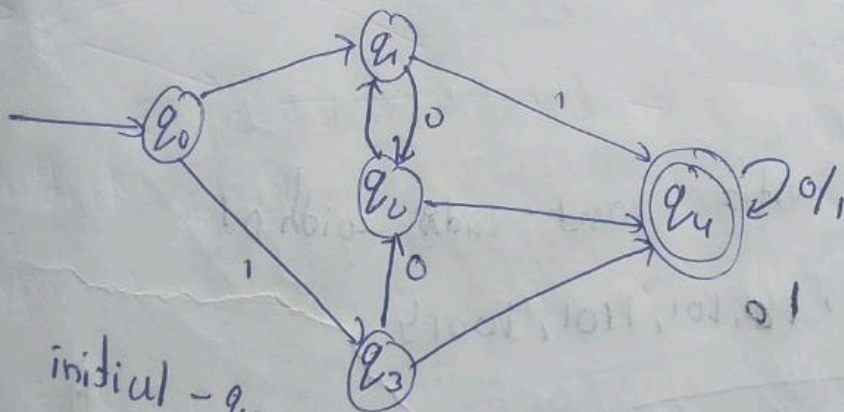
→ removed

	a	b
→ q ₀	q ₁	q ₀
q ₁	q ₁	q ₁
q ₂	q ₁	q ₂

Diagram :-



Given :-



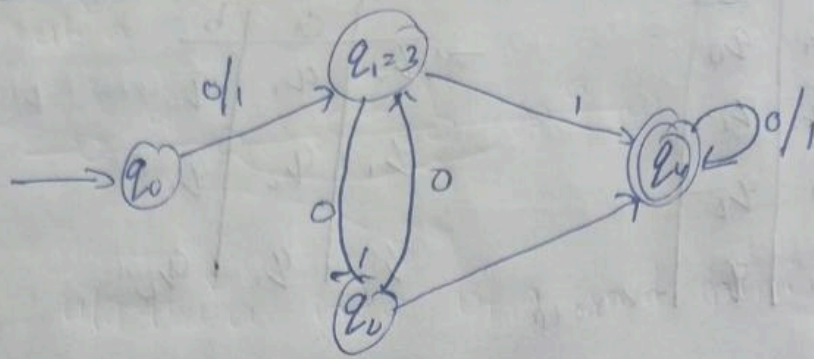
initial - q₀

final - q₄

There is no reachable from initial state

	a	b
→ q ₀	q ₁	q ₃
q ₁	q ₂	q ₄
q ₂	q ₁	q ₄
q ₃	q ₁	q ₄
*q ₄	q ₄	q ₄

	0	1
→ q ₀	q ₁	q ₃
q ₁ =3	q ₂	q ₄
q ₂	q ₁	q ₄
*q ₄	q ₄	q ₄



③

Given

set of all strings length '2'
over the a's and b's

$L = \{ \epsilon, a, b, ab, aa, bb, ba \}$

$(a+b)(a+b)$

④

given

state with 1 and ends with 01

$L = \{ \epsilon, 101, 1101, 1001 \}$

~~Reg. $\{ 01 \}^*$~~

~~$(01)^*$~~

~~$2 (1^* 01)$~~